LINEAR ALGEBRA

Comprehensive Study Notes

CO1 | RV University | First Year Computer Science

These notes cover the core Linear Algebra content from CO1: linear equations, systems, matrix representations, Gaussian elimination, LU decomposition, and matrix inverses. Every topic includes worked examples with matrices displayed in their standard bracket format, comparison tables, and exam-ready practice questions.

Module 1: Linear Equations & Systems

1.1 What is a Linear Equation?

Definition

A **linear equation** in n variables $x_1, x_2, \ldots, x_n$ has the form:

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where $a_1, a_2, \ldots, a_n$ and b are real constants, with at least one $a_i \neq 0$.

Examples vs Non-Examples

Linear ✓	NOT Linear ×
$y = mx + c$	$x^2 + 2x = 5$
$x + 3y = 7$	$y = \sin x$
$x_1 - 2x_2 - 3x_3 + 4x_4 - 9x_5 = 10$	$y = e^x$
$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$	$z = y^2 + 2z$

1.2 Solution of a Linear Equation

A **solution** to $a_1x_1 + \cdots + a_nx_n = b$ is a sequence $s_1, s_2, \ldots, s_n$ such that $a_1s_1 + \cdots + a_ns_n = b$. The set of all solutions is called the **solution set**.

1.3 System of Linear Equations

Definition

A **system of linear equations** is a finite set of linear equations sharing the same variables. A **solution** is a vector simultaneously satisfying every equation. The **solution set** is the set of all such vectors.

Example:

$$\begin{cases} 2x - y = 3 \\ x + y = 5 \end{cases}$$

Test $(1, -1)$: Eq. 1: $2(1) - (-1) = 3$ ✓ but Eq. 2: $1 + (-1) = 0 \neq 5$ ×. Answer: $(2, 1)$.

1.4 Graphical Interpretation

Case	System	Outcome	Type
(a)	$x - y = 1,\ x + y = 3$	Unique solution $[2, 1]$ — lines intersect	Consistent
(b)	$x - y = 2,\ 2x - 2y = 4$	Infinitely many — lines coincide	Consistent
(c)	$x - y = 1,\ x - y = 3$	No solution — parallel lines	Inconsistent

Key Theorem

A system of linear equations with real coefficients has **exactly one** of:

1. A **unique solution** (consistent)
2. **Infinitely many solutions** (consistent)
3. **No solution** (inconsistent)

Module 2: Matrix Representation

2.1 Matrix Form: $A\mathbf{x} = B$

An arbitrary system of m equations in n unknowns:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \qquad\qquad\qquad\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

can be written compactly as $A\mathbf{x} = B$, where:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad B = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

Example: The system $x + 2y + 3z = 6,\ 2x + 5y + 2z = 4,\ 6x - 3y + z = 2$ becomes:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 6 & -3 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \\ 2 \end{pmatrix}$$

2.2 Augmented Matrix $[A \,|\, B]$

Definition

The **augmented matrix** appends B as an extra column to A:

$$[A \,|\, B] = \left(\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{array}\right)$$

Example (same system as above):

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & 5 & 2 & 4 \\ 6 & -3 & 1 & 2 \end{array}\right)$$

2.3 Elementary Row Operations

#	Operation	Notation
1	Multiply row i by a non-zero constant c	$cR_i \to R_i$
2	Interchange rows i and j	$R_i \leftrightarrow R_j$
3	Add k times row j to row i	$R_i + kR_j \to R_i$

Module 3: Row Echelon Forms

3.1 Row Echelon Form (REF)

Definition

A matrix is in **Row Echelon Form (REF)** if:

1. All zero rows are at the bottom.
2. In any non-zero row, the first non-zero entry (**pivot**) is strictly to the right of the pivot in any row above it.
3. All entries below a pivot are zero.

REF Examples:

$$\begin{pmatrix} 1 & 4 & -3 & 7 \\ 0 & 1 & 6 & 2 \\ 0 & 0 & 1 & 5 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

3.2 Reduced Row Echelon Form (RREF)

Definition

A matrix is in **Reduced Row Echelon Form (RREF)** if it is in REF and additionally:

1. Every pivot is **1** (a *leading one*).
2. Every leading one is the **only non-zero entry** in its column (zeros above AND below).

RREF Examples:

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Feature	REF	RREF
Zero rows	At bottom	At bottom
Leading entry value	Any non-zero	Must be 1
Entries below pivot	All zero	All zero
Entries **above** pivot	Can be anything	**All zero**
Uniqueness	Not unique	**Unique**

Module 4: Gaussian Elimination

4.1 Algorithm

1. Write the augmented matrix $[A \mid B]$.

2. Apply row operations to reduce to REF.
3. Use **back-substitution** to solve (or continue to RREF for Gauss-Jordan).

4.2 Worked Example

Solve: $x + y + 2z = 9, \quad 2x + 4y - 3z = 1, \quad 3x + 6y - 5z = 0$

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right) \xrightarrow{R_2 \to R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 3 & 6 & -5 & 0 \end{array}\right) \xrightarrow{R_3 \to R_3 - 3R_1} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array}\right)$$

$$\xrightarrow{R_2 \to \frac{1}{2}R_2} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 3 & -11 & -27 \end{array}\right) \xrightarrow{R_3 \to R_3 - 3R_2} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & -\frac{1}{2} & -\frac{3}{2} \end{array}\right) \xrightarrow{R_3 \to -2R_3} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{array}\right)$$

Back-substitution: $z = 3$, then $y - \frac{7}{2}(3) = -\frac{17}{2} \Rightarrow y = 2$, then $x + 2 + 6 = 9 \Rightarrow x = 1$.

Solution: $x = 1, \quad y = 2, \quad z = 3$

4.3 Continuing to RREF (Gauss-Jordan)

Continuing from REF above:

$$\xrightarrow{R_1 \to R_1 - R_2} \left(\begin{array}{ccc|c} 1 & 0 & \frac{11}{2} & \frac{35}{2} \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{array}\right) \xrightarrow[R_2 \to R_2 + \frac{7}{2}R_3]{R_1 \to R_1 - \frac{11}{2}R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

Solution is read off directly: $x = 1,\ y = 2,\ z = 3$.

4.4 Another Example (Back-Substitution)

Solve: $-3x_1 + 2x_2 - x_3 = -1, \quad 6x_1 - 6x_2 + 7x_3 = -7, \quad 3x_1 - 4x_2 + 4x_3 = -6$

$$\begin{pmatrix} -3 & 2 & -1 \\ 6 & -6 & 7 \\ 3 & -4 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ -7 \\ -6 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} -3 & 2 & -1 & -1 \\ 6 & -6 & 7 & -7 \\ 3 & -4 & 4 & -6 \end{array}\right) \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 + 2R_1} \left(\begin{array}{ccc|c} -3 & 2 & -1 & -1 \\ 0 & -2 & 5 & -9 \\ 0 & -2 & 3 & -7 \end{array}\right) \xrightarrow{R_3 \to R_3 - R_2} \left(\begin{array}{ccc|c} -3 & 2 & -1 & -1 \\ 0 & -2 & 5 & -9 \\ 0 & 0 & -2 & 2 \end{array}\right)$$

Back-substitution: $-2x_3 = 2 \Rightarrow x_3 = -1$, then $-2x_2 + 5(-1) = -9 \Rightarrow x_2 = 2$, then $x_1 = 2$.

Solution: $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$

4.5 RREF Example

Find RREF of $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{pmatrix}$:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{pmatrix} \xrightarrow[R_3 \to R_3 - 6R_1]{R_2 \to R_2 - 4R_1} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -6 & -9 \\ 0 & -5 & -10 & -15 \end{pmatrix} \xrightarrow[R_3 \to R_3/(-5)]{R_2 \to R_2/(-3)} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{pmatrix}$$

$$\xrightarrow{R_3 \to R_3 - R_2} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \to R_1 - 2R_2} \begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \leftarrow \text{RREF}$$

Pivot columns: C_1, C_2. Free variables: x_3, x_4.

Module 5: Matrix Inverse via Row Reduction

5.1 Definition

Definition

The **inverse** A^{-1} of a square matrix A is the unique matrix satisfying $AA^{-1} = A^{-1}A = I$. It exists iff $\det(A) \neq 0$ (non-singular).

5.2 Algorithm

Form $[A \,|\, I]$ and row-reduce the left side to I; the right side becomes A^{-1}:

$$[A \,|\, I] \;\longrightarrow\; [I \,|\, A^{-1}]$$

5.3 Worked Example

Find A^{-1} for $A = \begin{pmatrix} -3 & 2 & -1 \\ 6 & -6 & 7 \\ 3 & -4 & 4 \end{pmatrix}$

$$\left(\begin{array}{ccc|ccc} -3 & 2 & -1 & 1 & 0 & 0 \\ 6 & -6 & 7 & 0 & 1 & 0 \\ 3 & -4 & 4 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 + 2R_1} \left(\begin{array}{ccc|ccc} -3 & 2 & -1 & 1 & 0 & 0 \\ 0 & -2 & 5 & 2 & 1 & 0 \\ 0 & -2 & 3 & 1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left(\begin{array}{ccc|ccc} -3 & 2 & -1 & 1 & 0 & 0 \\ 0 & -2 & 5 & 2 & 1 & 0 \\ 0 & 0 & -2 & -1 & -1 & 1 \end{array}\right) \xrightarrow{R_1 \to R_1 + R_2} \left(\begin{array}{ccc|ccc} -3 & 0 & 4 & 3 & 1 & 0 \\ 0 & -2 & 5 & 2 & 1 & 0 \\ 0 & 0 & -2 & -1 & -1 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \to R_1 + 2R_3,\ R_2 \to R_2 + \frac{5}{2}R_3} \left(\begin{array}{ccc|ccc} -3 & 0 & 0 & 1 & -1 & 2 \\ 0 & -2 & 0 & -\frac{1}{2} & -\frac{3}{2} & \frac{5}{2} \\ 0 & 0 & -2 & -1 & -1 & 1 \end{array}\right) \xrightarrow{\substack{R_1 \to R_1/(-3) \\ R_2 \to R_2/(-2) \\ R_3 \to R_3/(-2)}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{3} & \frac{1}{3} & -\frac{2}{3} \\ 0 & 1 & 0 & \frac{1}{4} & \frac{3}{4} & -\frac{5}{4} \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array}\right)$$

$$A^{-1} = \begin{pmatrix} -\frac{1}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{1}{4} & \frac{3}{4} & -\frac{5}{4} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

Module 6: LU Decomposition

6.1 What is LU Decomposition?

Definition

LU decomposition factors a square matrix A as:

$$A = LU$$

where L is **lower triangular** and U is **upper triangular**. General 3×3 form (with 1s on diagonal of L):

$$L = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}$$

Note: LU decomposition is NOT unique — it depends on the chosen convention for L and U.

6.2 Solving $A\mathbf{x} = \mathbf{b}$ via LU

Since $A = LU$, the system $A\mathbf{x} = \mathbf{b}$ becomes $LU\mathbf{x} = \mathbf{b}$. Let $\mathbf{y} = U\mathbf{x}$:

1. **Forward substitution:** Solve $L\mathbf{y} = \mathbf{b}$ for $\mathbf{y}$ (easy — L is lower triangular).
2. **Back substitution:** Solve $U\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$ (easy — U is upper triangular).

6.3 Finding L and U via Row Operations

When performing Gaussian elimination $R_i \to R_i - k \cdot R_j$ (with $i > j$), the multiplier k becomes the entry l_{ij}. After all operations, the resulting matrix is U.

$$\text{Row op } R_i \to R_i - k\,R_j \implies l_{ij} = k$$

6.4 Worked Example — 3×3 LU

Find L and U for $A = \begin{pmatrix} 2 & 1 & 3 \\ 4 & -1 & 3 \\ -2 & 5 & 5 \end{pmatrix}$

$R_2 \to R_2 - 2R_1$ ($l_{21} = 2$):

$$\begin{pmatrix} 2 & 1 & 3 \\ 0 & -3 & -3 \\ -2 & 5 & 5 \end{pmatrix}$$

$R_3 \to R_3 + R_1$ ($l_{31} = -1$):

$$\begin{pmatrix} 2 & 1 & 3 \\ 0 & -3 & -3 \\ 0 & 6 & 8 \end{pmatrix}$$

$R_3 \to R_3 + 2R_2$ ($l_{32} = -2$):

$$\begin{pmatrix} 2 & 1 & 3 \\ 0 & -3 & -3 \\ 0 & 0 & 2 \end{pmatrix} = U$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -2 & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -3 & -3 \\ 0 & 0 & 2 \end{pmatrix}$$

6.5 Worked Example — 4×4 LU

Find L and U for:

$$A = \begin{pmatrix} 3 & 1 & 3 & -4 \\ 6 & 4 & 8 & -10 \\ 9 & 2 & 5 & -1 \\ -9 & 5 & -2 & -4 \end{pmatrix}$$

Row operations with multipliers:

$R_2 \to R_2 - 2R_1$ ($l_{21} = 2$), $R_3 \to R_3 - 3R_1$ ($l_{31} = 3$), $R_4 \to R_4 + 3R_1$ ($l_{41} = -3$):

$$\begin{pmatrix} 3 & 1 & 3 & -4 \\ 0 & 2 & 2 & -2 \\ 0 & -1 & -4 & 11 \\ 0 & 8 & 7 & -16 \end{pmatrix}$$

$R_3 \to R_3 + \frac{1}{2}R_2$ ($l_{32} = -\frac{1}{2}$), $R_4 \to R_4 - 4R_2$ ($l_{42} = 4$):

$$\begin{pmatrix} 3 & 1 & 3 & -4 \\ 0 & 2 & 2 & -2 \\ 0 & 0 & -3 & 10 \\ 0 & 0 & -1 & -8 \end{pmatrix}$$

$R_4 \to R_4 - \frac{1}{3}R_3$ ($l_{43} = \frac{1}{3}$):

$$\begin{pmatrix} 3 & 1 & 3 & -4 \\ 0 & 2 & 2 & -2 \\ 0 & 0 & -3 & 10 \\ 0 & 0 & 0 & -\frac{34}{3} \end{pmatrix} = U$$

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & -\frac{1}{2} & 1 & 0 \\ -3 & 4 & \frac{1}{3} & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} 3 & 1 & 3 & -4 \\ 0 & 2 & 2 & -2 \\ 0 & 0 & -3 & 10 \\ 0 & 0 & 0 & -\frac{34}{3} \end{pmatrix}$$

6.6 LU Solve — Full Example

Solve $A\mathbf{x} = \mathbf{b}$ where $A = \begin{pmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -3 \\ 3 \\ 2 \end{pmatrix}$

LU factorisation:

$$A = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{pmatrix}}_{L} \underbrace{\begin{pmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{pmatrix}}_{U}$$

Step 1 — Forward sub $L\mathbf{y} = \mathbf{b}$:

$$\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} -3 \\ 3 \\ 2 \end{pmatrix} \quad \Rightarrow \quad y_1 = -3, \quad y_2 = 0, \quad y_3 = 8$$

Step 2 — Back sub $U\mathbf{x} = \mathbf{y}$:

$$\begin{pmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \\ 8 \end{pmatrix} \quad \Rightarrow \quad x_3 = -8, \quad x_2 = 4, \quad x_1 = 3$$

$$\textbf{Solution: } \mathbf{x} = \begin{pmatrix} 3 \\ 4 \\ -8 \end{pmatrix}$$

Module 7: Methods Summary

Method	Type	Key Feature
Gaussian Elimination	Direct	Reduce to REF, then back-substitute
Gauss-Jordan	Direct	Reduce to RREF — no back-sub needed
LU Decomposition	Direct	Factor $A = LU$; efficient for repeated solves
Matrix Inversion	Direct	$\mathbf{x} = A^{-1}\mathbf{b}$; requires invertible A
Jacobi / Gauss-Seidel	Iterative	Converges for diagonally dominant A
Cramer's Rule	Direct	Uses determinants; impractical for large n

Module 8: Applications

8.1 Chemical Equation Balancing

Balance: $x_1\,CH_4 + x_2\,O_2 \to x_3\,CO_2 + x_4\,H_2O$
Atom conservation gives the homogeneous system:

$$\begin{pmatrix} 1 & 0 & -1 & 0 \\ 4 & 0 & 0 & -2 \\ 0 & 2 & -2 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \mathbf{0}$$

Set $x_1 = 1$ (free variable): $x_1 = 1,\ x_2 = 2,\ x_3 = 1,\ x_4 = 2$.
Balanced: $CH_4 + 2O_2 \to CO_2 + 2H_2O$ ✓

8.2 Polynomial Curve Fitting

Find cubic $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$ through $(1,3), (2,-2), (3,-5), (4,0)$:

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 64 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ -5 \\ 0 \end{pmatrix}$$

Solving via augmented matrix gives $p(x) = 4 + 3x - 5x^2 + x^3$.

Module 9: Exam Preparation

9.1 Top 15 Testable Facts

1. A linear equation has variables to the **first power only** — no x^2, xy, $\sin x$, e^x.
2. A system has exactly **0, 1, or** ∞ solutions.

3. The augmented matrix $[A|B]$ encodes the full system $A\mathbf{x} = B$.
4. Three elementary row operations: **scale**, **swap**, **add multiple**.
5. REF: zero rows at bottom, leading entries step rightward, zeros below pivots.
6. RREF: REF + leading 1s + zeros **above and below** each pivot.
7. The RREF of a matrix is **unique**; REF is **not unique**.
8. Gaussian elimination: reduce $[A|B]$ to REF, then back-substitute.
9. Gauss-Jordan: reduce to RREF — read solution directly.
10. A^{-1} exists iff $\det(A) \neq 0$. Find via $[A|I] \to [I|A^{-1}]$.
11. LU: $A = LU$ where L is lower, U is upper triangular.
12. LU solve: $L\mathbf{y} = \mathbf{b}$ (forward), then $U\mathbf{x} = \mathbf{y}$ (backward).
13. LU is **not unique** — depends on the chosen convention.
14. Multiplier rule: $R_i \to R_i - kR_j \;\Rightarrow\; l_{ij} = k$.
15. Pivot column $\to$ basic variable. Non-pivot column $\to$ free variable.

9.2 Common Mistakes

Common Mistake

- Forgetting to check ALL equations when verifying a solution.
- Confusing REF and RREF — entries above a pivot can be non-zero in REF but must be zero in RREF.
- Wrong sign for LU multiplier — if $R_i \to R_i - kR_j$, then $l_{ij} = +k$ (not $-k$).
- Treating LU as unique — it is not.
- Mixing up forward (uses L) and backward (uses U) substitution.
- Calling a no-solution system "consistent" — it is **inconsistent**.
- Applying row ops only to A, not the full augmented matrix $[A|B]$.

9.3 Practice Questions

MCQs

Q1. Which is NOT a linear equation?
(A) $2x + 3y = 7$ (B) $x_1 - 4x_2 + x_3 = 0$ **(C)** $y = e^x$ (D) $x - y + 2z = 5$

Q2. In LU decomposition, what is done *first* to solve $A\mathbf{x} = \mathbf{b}$?
(A) Solve $U\mathbf{x} = \mathbf{y}$ (B) Compute A^{-1} **(C) Solve $L\mathbf{y} = \mathbf{b}$** (D) Reduce A to RREF

Q3. The RREF of a matrix is:
(A) Not unique **(B) Unique** (C) Same as REF (D) Always the identity

Q4. Row op $R_3 \to R_3 - 5R_1$ gives LU entry:
(A) -5 **(B)** 5 (C) 1/5 (D) 0

Descriptive Questions

1. Solve using Gaussian elimination: $x + y + 2z = 9,\ 2x + 4y - 3z = 1,\ 3x + 6y - 5z = 0.$
2. Find the RREF of $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{pmatrix}$ and identify all pivot columns.

3. Find L and U for $A = \begin{pmatrix} 2 & 1 & 3 \\ 4 & -1 & 3 \\ -2 & 5 & 5 \end{pmatrix}$ such that $LU = A$.

4. Using LU, solve $A\mathbf{x} = \mathbf{b}$ for $A = \begin{pmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -3 \\ 3 \\ 2 \end{pmatrix}$.

5. Compute A^{-1} for $A = \begin{pmatrix} -3 & 2 & -1 \\ 6 & -6 & 7 \\ 3 & -4 & 4 \end{pmatrix}$.

6. Show that the system $x - y = 1,\ x - y = 3$ is inconsistent.

7. Balance $CH_4 + O_2 \rightarrow CO_2 + H_2O$ using a system of linear equations.

Cheat Sheet — One-Page Quick Reference

Key Definitions

- **Linear eq.**: $a_1x_1 + \cdots + a_nx_n = b$, first power only
- **Consistent**: ≥ 1 solution; **Inconsistent**: none
- **Augmented matrix**: $[A|B]$
- **REF**: pivots step right, zeros below
- **RREF**: REF + leading 1s + zeros above & below
- **Pivot**: first non-zero entry in a row
- **Free variable**: corresponds to non-pivot column
- **LU**: $A = LU$, L lower, U upper triangular
- A^{-1}: exists iff $\det(A) \neq 0$

Algorithms

- **Gauss**: $[A|B] \to$ REF $\to$ back-sub
- **Gauss-Jordan**: $[A|B] \to$ RREF $\to$ read off
- **Inverse**: $[A|I] \to$ RREF $\to [I|A^{-1}]$
- **LU find**: row-reduce $A \to U$; l_{ij} = multiplier
- **LU solve**: $L\mathbf{y} = \mathbf{b}$ then $U\mathbf{x} = \mathbf{y}$

Row Operations

- $cR_i \to R_i$: scale row i by $c \neq 0$
- $R_i \leftrightarrow R_j$: swap rows
- $R_i + kR_j \to R_i$: add $k\times$ row j to row i
- If $R_i \to R_i - kR_j$: $l_{ij} = k$

LU Structure (3×3)

$$L = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix} \quad U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}$$

Types of Solutions

$$\left(\begin{array}{cc|c} 1 & 0 & a \\ 0 & 1 & b \end{array}\right) \to \text{unique}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & * & a \\ 0 & 0 & 0 & 0 \end{array}\right) \to \infty \text{ solutions}$$

$$\left(\begin{array}{cc|c} 1 & 0 & a \\ 0 & 0 & b \end{array}\right)_{b\neq 0 \to \text{inconsistent}}$$

Mnemonics

- **RREF** = “Really Reduced” — 1s on pivots, zeros everywhere in pivot columns
- **LU solve**: “*L*eft (forward), *U*p (backward)”
- **Multiplier sign**: subtract $k\times$ row $\Rightarrow$ store $+k$ in L

References

- Lay et al., *Linear Algebra and Its Applications*, 5th ed., Pearson, 2016.
- Anton & Rorres, *Elementary Linear Algebra*, 11th ed., Wiley, 2013.
- RV University CO1 Class Notes (primary source).